

Biology Ch 8, Part 1: Nutrition aur Transport in Plants

Aasan Notes for Seemab · kitab ke page 113 se 117 tak · 8.1, 8.2, 8.2.1, 8.2.2, 8.2.3

Pehle ek sawal. Ek poplar ka darakht roz zameen se kareeban 400 litre paani kheenchta hai. Ab andaza lagao: us 400 litre mein se kitna paani darakht apne kaam mein laata hai? Aadha? Choutha hissa? Jawab sunn kar aap hairan reh jayengi, aur wo is note ke aakhir mein hai.

1. Nutrition in Plants (p113)

Kitab ki definition

Nutrient: "Any substance that provides necessary elements to the organism for growth and metabolism is called nutrient."

Nutrition: the utilization of nutrients in growth or various metabolic activities of the body.

Plants **autotrophs** hain: wo inorganic nutrients jaise paani, carbon dioxide aur kuch minerals environment se lete hain aur unhein khud **organic compounds** mein badal lete hain. Yani apna khana khud banate hain.

8.1.1 Types of plant nutrients

Kitab kehti hai **sixteen elements** plant growth ke liye essential hain. In mein se **nau (9)** bari miqdar mein chahiyein.

- **Macronutrients** (0.05% dry weight se **zyada**): carbon, hydrogen, oxygen, nitrogen, phosphorus, potassium, sulphur, calcium, magnesium
- **Micronutrients** (0.05% dry weight se **kam**): iron, boron, manganese, copper, molybdenum, chlorine, zinc

Farq yaad rakhne ka tareeqa

Macro = bari miqdar, Micro = traces. Line **0.05% dry weight** par hai. Dono list mein 9 aur 7 elements hain, total 16. Aur yaad rakhein: **har nutrient essential hai**, chahe kitni hi kam miqdar mein ho. Kam zaroorat ka matlab kam ahmiyat nahin.

Do nutrients jo imtihan mein aksar aate hain

Nitrogen (protein synthesis): plant ise **nitrates (NO_3^-)** ki shakl mein soil se leti hai. Nitrates **dead organic matter** ke decomposition se bante hain. Nitrogen ki kami ho to **nitrogenous fertilizers** daale jate hain. Plant body mein nitrogenous compounds se **amino acids, nucleotides aur chlorophyll** banta hai, aur amino acids se protein. Kami ki sab se numayan alamat: **severe growth retardation.**

Magnesium (chlorophyll formation): plant ise **inorganic Mg^{2+} ions** ki shakl mein leti hai. Source: soil mein **dolomitic limestone**, ya fertilizer **Magnesium sulphate (MgSO_4)**. Magnesium chlorophyll banane mein istemal hota hai, is liye kami se **leaves peeli** ho jati hain (chlorosis).

2. Transport in Plants (p114, p115)

8.2.1 Need of Transport, teen levels

1st level: material andar se bahar aur bahar se andar, taake har cell ko nutrients milein.

2nd level: jahan material banta hai wahan se jahan istemal hota hai, jaise **jarr se patton tak.**

3rd level: plant body ke andar se bahar, kyunke kuch compounds bekar (waste) hote hain.

Root aur root hairs (8.2.2). Root plant ka **underground** hissa hai aur mineral nutrients aur paani lene ka organ hai. Is ka surface area **branching system** aur **root hairs** se bara ho jata hai, jo absorption ke liye behtareen shakl hai. **Dicot** plants mein **tap root** hota hai (ek mota primary root), aur **monocot** plants mein **fibrous adventitious roots.**

🧠 Root ke andar ki tarteeb, bahar se andar

Epidermis → Cortex → Endodermis → Pericycle → Xylem. Yehi tarteeb paani ka raasta hai. Endodermis ki deewaron par **suberin** ki pattiyan hoti hain jinhein **Casparian Strips** kehte hain.

⚡ 3. Uptake: Passive aur Active (p115, p116)

	Passive transport	Active transport
Energy	Koi energy nahin	ATP ki energy lagti hai
Direction	Concentration gradient ke saath	Gradient ke khilaf
Kaam kab	Jab bahar zyada, andar kam	Jab root cells mein pehle hi zyada ho
Kitab ke lafz	diffusion, facilitated diffusion, osmosis	selective, relies on respiration

📖 Facilitated diffusion

"Proteins within the cell membrane act as carrier molecules to help move nutrients across the membrane." Ye carrier proteins epidermal aur doosray root cells ki cell membranes mein hote hain.

📖 Osmosis, kitab ki apni definition

"The movement of water molecules through a partially permeable membrane from an area where they are more concentrated to an area where they are less concentrated."

Paani cell ke **andar** jaye to **endosmosis**, cell se **bahar** jaye to **exosmosis**.

⚠️ Ye farq imtihan mein poochha jata hai

Root ki epidermal cells mein **cell wall** paani aur minerals ko azadi se guzarne deti hai, lekin **cell membrane selectively permeable** hai, yani wo sirf kuch cheezon ko guzarne deti hai. Cell wall aur cell membrane ko mat milayein.

💧 4. Transpiration (p117)

📖 Kitab ki definition

Jarr se liya hua paani plant ke **aerial parts** tak jata hai, aur us mein se **taqreeban 99%** vapours ki shakl mein hawa mein chala jata hai. "This loss is called transpiration."

Qism	Kitna	Kahan se
Stomatal transpiration	90%	Stomata, jo zyada tar patton mein hote hain
Cuticular transpiration	7 se 9%	Cuticle, yani poori body ki upri satah
Lenticular transpiration	3% se kam	Lenticels, tane par nishan jaise hisse

🧠 Transpiration ko kitab "necessary evil" kyun kehti hai

Faide: plant ko **cooling** deti hai; paani aur ghule hue minerals absorb karne mein madad deti hai; aur unhein **upar ki taraf** patton tak le jati hai. Is ke baghair plant zinda nahin reh sakti.

Nuqsan: itna paani zaya hota hai ke agar transpiration bay-qaboo ho jaye to plant **wilting** (loss of **turgor**) ka shikar ho kar mar bhi sakti hai.

Dono baatein ek saath likhna, tabhi jawab poora hota hai.

Leaf surface area aur stomata ka asar. Transpiration zyada tar **stomata** se hoti hai, jo patte ki satah par hote hain. Is liye jin plants ke **patton ki satah bari** ho aur stomata zyada hon, un mein transpiration ki rate **zyada** hoti hai; aur jin ke patte **tang aur chhote** hon un mein **kam**. Isi tarah

din mein raat se zyada transpiration hoti hai, kyunke stomata din mein khule rehte hain aur raat ko taqreeban band ho jate hain.

Ab shuru wala sawal

Darakht 400 litre mein se apne kaam mein laata hai sirf **ek litre ke qareeb**, kyunke **99% paani transpiration mein chala jata hai**. Ajeeb lagta hai na? Lekin wohi 99% wo pump hai jo minerals ko jarr se patton tak kheench kar le jata hai. Isi liye kitab ise **necessary evil** kehti hai: nuqsan lagta hai, magar us ke baghair plant zinda nahin reh sakti.

Quiz

Ye teeno sawal tumhare Biology paper mein bhi aayenge, is liye inhein achhi tarah karo.

1. Define a nutrient in the words of your book. Then name the **nine macronutrients** and state the percentage of dry weight that separates macronutrients from micronutrients.
2. (i) In which form does a plant take nitrogen from the soil, and which three compounds are made from it in the plant body? (ii) Name the deficiency symptom the book gives for nitrogen, and the one it gives for magnesium.
3. Write the three types of transpiration with the percentage of water lost in each, then explain in two sentences why the book calls transpiration a "necessary evil".